Mechanism-Oriented Comparison of Time-Series Clustering Similarity Measures across Noise, Misalignment, and Length

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1 Introduction

Time-series clustering relies fundamentally on how similarity between sequences is defined. Different similarity measures encode distinct assumptions about alignment, temporal variation, and signal structure. *Lock-step* measures such as Euclidean distance (ED) compare time points one-to-one; *elastic* measures such as Dynamic Time Warping (DTW) [1] and Move–Split–Merge (MSM) [2] permit local temporal warping; *sliding* measures such as Shape-Based Distance (SBD) [3] emphasise shape similarity under global phase shifts via normalised cross-correlation; and *distributional kernel* methods such as the Isolation Distributional Kernel (IDK) [8] compare subsequence distributions without explicit point-to-point alignment.

Extensive benchmark studies have compared time-series distances and clustering algorithms across large archives [11, 6], yet they predominantly report aggregate performance on clean datasets. This leaves a *mechanistic* question unanswered: how do noise, temporal misalignment, and sequence length individually affect the relative performance of different similarity paradigms? The question is practically important because a measure that excels on well-aligned data may collapse under phase shifts, whereas a more invariant measure may discard temporal information that certain classes require.

This paper addresses this gap through a unified time-series clustering framework that compares ED, DTW, MSM, SBD, and IDK under both real-data benchmarking and controlled perturbation experiments. We adopt a k-medoids pipeline with a shared evaluation protocol across 18 UCR datasets [4] and apply three systematic perturbations—additive Gaussian noise, random temporal shift, and truncation-based length reduction—on representative subsets. Our goal is *not* to crown a single universal winner, but to explain under what data-generating conditions each similarity paradigm is most appropriate.

The contributions of this work are:

1. A fair, mechanism-oriented comparison of five representative similarity measures spanning the lock-step, elastic, sliding, and distributional paradigms.

2. A unified k-medoids clustering pipeline with shared result schema, fixed hyperparameters, and 10-seed repetition for statistical rigour.

3. A controlled perturbation study that isolates the effects of noise, misalignment, and temporal resolution on clustering quality, producing interpretable degradation curves.

4. Mechanistic explanations linking each measure's invariance properties to its empirical robustness profile.

2 Related Work

2.1 Benchmark Studies of Time-Series Distances

Large-scale comparisons of time-series distance functions have established taxonomies that categorise measures by their alignment strategy. Paparrizos et al. [11] evaluated over 70 distance functions across the UCR archive, introducing a principled normalisation framework and statistical ranking methodology. Their findings demonstrated that no single distance dominates universally and that z-normalisation is a prerequisite for fair comparison—a practice we adopt throughout. Bagnall et al. [4] provided the seminal classification benchmark across 85 UCR datasets, establishing DTW with a learned warping window as the baseline to beat. While these studies provide comprehensive cross-paradigm rankings, they are oriented toward classification rather than clustering and do not systematically isolate the effect of individual data perturbations on distance performance.

2.2 Time-Series Clustering Benchmarks

Javed et al. [6] benchmarked time-series clustering methods using ED, DTW, and shape-based distances with various clustering algorithms, confirming that the choice of distance function matters more than the choice of clustering algorithm for many datasets. Paparrizos and Gravano [3] demonstrated that SBD combined with centroid refinement (k-Shape) achieves competitive clustering quality with substantially lower computational cost than DTW-based approaches. These works establish ED, DTW, and SBD as essential baselines for any clustering comparison. However, they do not include IDK—a fundamentally different distributional paradigm—nor do they apply controlled noise, shift, or length sweeps to characterise robustness mechanisms.

2.3 Elastic Distances for Clustering

Beyond DTW, the elastic distance family includes measures such as MSM [2], Time Warp Edit Distance (TWE), and Edit Distance on Real Sequences (EDR). Holder and Bagnall [9] demonstrated that MSM can outperform DTW for clustering when the distance geometry aligns with the clustering objective, particularly in datasets where amplitude changes carry discriminative information. Their analysis supports the use of k-medoids (rather than k-means) as the clustering algorithm, since medoid-based methods only require pairwise distances and do not assume a meaningful averaging operation in the original space. This motivates our inclusion of MSM alongside DTW and our adoption of k-medoids as the unified clustering backbone.

2.4 Distributional and Kernel-Based Approaches

Distributional approaches represent time series through summary statistics of their subsequence structure rather than explicit point-to-point alignment. The Isolation Distributional Kernel (IDK) [8] maps each instance into a feature space derived from isolation mechanisms, yielding a positive semi-definite kernel. Gong et al. [10] applied IDK to time-series clustering via CTDS, demonstrating competitive performance against shape-based methods on selected UCR datasets. However, IDK has not been systematically compared with ED, DTW, MSM, and SBD within a unified k-medoids framework, nor has its behaviour under controlled perturbations been characterised. Our work frames IDK as representing a *distributional paradigm*—not merely an additional distance entry—and examines whether its lack of explicit temporal alignment confers robustness to noise and misalignment at the cost of temporal-order sensitivity.

2.5 Soft-DTW and Differentiable Alignment

Cuturi and Blondel [5] introduced Soft-DTW, a differentiable relaxation of DTW that enables gradient-based optimisation of barycentres and has been widely adopted in deep learning pipelines for time series. While Soft-DTW is not included as a primary measure in our comparison (it is designed as a loss function rather than a distance for medoid selection), its boundary-handling conventions—particularly edge padding for shift experiments—inform our perturbation protocol design (Section 3.5).

3 Methodology

3.1 Overview

We evaluate univariate whole time-series clustering using a unified k-medoids pipeline. Each dataset is represented as a matrix $X \in \mathbb{R}^{n \times L}$, where n is the number of instances and L is the series length. The number of clusters k is set to the ground-truth number of classes to ensure fair evaluation; class labels are used *only* for computing external validity indices (ARI and NMI) and never participate in the clustering process. All series are z-normalised (zero mean, unit variance) prior to similarity computation, following the UCR-standard preprocessing protocol [7].

The experimental design consists of two complementary parts:

- **Part 1: Real-data benchmark.** We evaluate all five measures on 18 selected UCR datasets covering a range of series lengths ($L \in [24, 1460]$), class counts ($k \in [2, 10]$), and application domains.
- **Part 2: Controlled perturbation study.** We apply three systematic transformations—noise, misalignment, and length reduction—to representative datasets and trace the resulting degradation of clustering quality.

3.2 Similarity Measures

We compare five measures spanning four similarity paradigms:

1. **Euclidean Distance (ED)** — *lock-step.* Point-wise ℓ_2 distance requiring exact temporal alignment: $d_{\mathrm{ED}}(x, y) = \sqrt{\sum_{t=1}^{L}(x_t - y_t)^2}$.
2. **Dynamic Time Warping (DTW)** — *elastic.* Finds an optimal alignment path under a Sakoe–Chiba band constraint with window $w = \lfloor 0.1 \cdot L \rfloor$ [6]: $d_{\mathrm{DTW}}(x, y) = \min_{\pi \in \mathcal{A}_w} \sqrt{\sum_{(i,j) \in \pi}(x_i - y_j)^2}$.
3. **Move–Split–Merge (MSM)** — *elastic/edit.* Extends elastic alignment with split and merge edit operations at fixed cost $c = 1.0$ [2, 9].
4. **Shape-Based Distance (SBD)** — *sliding.* Defined as $d_{\mathrm{SBD}}(x, y) = 1 - \max_w \mathrm{NCC}_w(x, y)$, where NCC_w denotes the normalised cross-correlation at lag w [3]. SBD is by construction invariant to circular shifts of its inputs.
5. **Isolation Distributional Kernel (IDK)** — *distributional.* IDK [8] produces a positive semi-definite kernel $K(x, y) \in [0, 1]$. To integrate IDK into the k-medoids framework, we convert it to a metric distance via the kernel-induced transformation:
$$d_{\mathrm{IDK}}(x, y) = \sqrt{K(x, x) + K(y, y) - 2\,K(x, y)}. \quad (1)$$
Because IDK feature maps are L_2-normalised (i.e., $K(x, x) = 1$), this simplifies to $d_{\mathrm{IDK}}(x, y) = \sqrt{2 - 2\,K(x, y)}$. This formulation is mathematically equivalent to the Euclidean distance implicitly computed by CTDS [10] when applying k-means on L_2-normalised IDK feature mean maps, ensuring consistency with prior work while adapting to our medoid-based pipeline.

No hyperparameter tuning was performed; all measures use default or literature-recommended settings. This is a deliberate fairness constraint: the comparison targets the *paradigm-level* properties of each measure, not the outcome of measure-specific optimisation.

3.3 Datasets

We select 18 datasets from the UCR Time Series Archive [4], summarised in Table 1. The selection criteria are: (i) coverage of series lengths from 24 to 1460, (ii) inclusion of binary and multi-class problems (k from 2 to 10), and (iii) representation of diverse domains including synthetic, sensor, medical, motion, shape, spectroscopy, and device signals.

Table 1: Selected UCR datasets for Part 1 benchmark.

#	Dataset	L	k	n	Domain
1	Chinatown	24	2	363	Traffic
2	SyntheticControl	60	6	600	Synthetic
3	MoteStrain	84	2	1272	Sensor
4	ECG200	96	2	200	Medical
5	CBF	128	3	930	Synthetic
6	TwoPatterns	128	4	5000	Synthetic
7	ECGFiveDays	136	2	884	Medical
8	Plane	144	7	210	Sensor
9	GunPoint	150	2	200	Motion
10	Wine	234	2	111	Spectroscopy
11	ArrowHead	251	3	211	Shape
12	Trace	275	4	200	Synthetic
13	Coffee	286	2	56	Spectroscopy
14	DiatomSizeReduction	345	4	322	Image
15	Symbols	398	6	1020	Shape
16	OSULeaf	427	6	442	Shape
17	Computers	720	2	500	Device
18	ACSF1	1460	10	200	Device

For datasets where full pairwise computation is prohibitively expensive (TwoPatterns, Symbols, Computers), we apply stratified sampling (equal instances per class) with a fixed subsample seed to ensure all measures are evaluated on identical subsets.

3.4 Clustering Protocol

For each dataset–measure combination, we:

1. Compute the full $n \times n$ pairwise distance matrix.
2. Run the PAM (Partitioning Around Medoids) algorithm [12] with k equal to the ground-truth class count, using the `alternate` heuristic for initialisation.
3. Repeat with 10 independent clustering seeds ($s = 1, 2, \ldots, 10$) to capture initialisation variance.
4. Report the Adjusted Rand Index (ARI) and Normalised Mutual Information (NMI) averaged over seeds, with standard deviations.

Cross-measure statistical comparison uses the Friedman test over datasets, with average ranks as the primary summary statistic. This follows established practice in time-series benchmarking [4, 11].

3.5 Perturbation Study

The perturbation study uses three representative datasets: CBF ($L = 128$, $k = 3$), Trace ($L = 275$, $k = 4$), and ECG200 ($L = 96$, $k = 2$). These were selected for their clear class structures, moderate sizes (enabling dense perturbation $\times$ seed grids), and coverage of synthetic and real signals.

All randomness is generated through `numpy.random.Generator` (PCG64) seeded from a single `SeedSequence`; per-cell sub-streams are obtained via `SeedSequence.spawn`, ensuring independence and bit-exact reproducibility across runs.

3.5.1 Noise Perturbation

Gaussian noise is injected as:

$$x'(t) = x(t) + \varepsilon(t), \quad \varepsilon(t) \sim \mathcal{N}(0, \sigma^2), \tag{2}$$

where $\sigma = \ell \cdot \mathrm{std}(x)$ and ℓ is the noise level swept over $\{0.0, 0.1, 0.2, 0.4, 0.8\}$. Because all series are z-normalised prior to perturbation ($\mathrm{std}(x) = 1$), σ numerically equals ℓ. The relative formulation ensures cross-dataset comparability: the level $\ell = 0.0$ serves as the unperturbed baseline, while $\ell = 0.8$ (SNR ≈ 2 dB) probes whether each metric retains residual discriminative power when noise approaches the signal's own variance.

3.5.2 Misalignment Perturbation

Misalignment is introduced by shifting each series independently by an integer amount $s_i \sim \mathrm{Uniform}\{-S, +S\}$, where the maximum shift S corresponds to $\{0\%, 5\%, 10\%, 20\%, 30\%\}$ of the series length L. Per-instance shifts are required because a globally uniform shift leaves all pairwise Euclidean distances invariant and therefore fails to constitute a meaningful perturbation.

After shifting, vacated positions are filled by *edge replication* (the first or last observed value, depending on shift direction). Edge padding is chosen as the default because:

1. The target datasets are not periodic—circular wrap-around would graft semantically incompatible regions.
2. SBD is by construction invariant under circular shifts; a circular-default protocol would trivially yield zero degradation, conflating metric definition with empirical robustness.

A circular-shift ablation is reported separately to characterise SBD's intrinsic translation invariance as a distinct finding.

3.5.3 Length Perturbation

Length perturbation is performed by symmetrically truncating each series from both ends to $\lceil L \cdot f \rceil$ samples (with a floor of 2), then resampling back to the original length L via FFT-based polyphase interpolation (`scipy.signal.resample`). Truncation fractions are swept over $f \in \{1.0, 0.9, 0.75, 0.5, 0.25\}$, where $f = 1.0$ is the unperturbed control. Zero-padding is deliberately excluded: on z-normalised signals it introduces a boundary step discontinuity that confounds length change with edge-artifact injection. Two-sided truncation preserves the signal's temporal centre of mass, ensuring that the diagnostic waveform region (e.g., the QRS complex in ECG200 or the bell shape in CBF) is not systematically destroyed.

3.6 Expected Mechanism Table

Table 2 summarises the invariance properties and expected perturbation behaviour of each measure, which guide our interpretation of the degradation curves.

These hypotheses are *a priori* and will be confronted with empirical degradation curves in Section 4. Our overarching claim is not that any single measure is universally superior, but rather:

Table 2: Similarity paradigms and expected perturbation responses.

Measure	Mechanism	Expected behaviour
ED	Pointwise lock-step	Strong on aligned data; degrades under shift.
DTW	Local elastic warping	Robust to local timing changes; may overfit noise.
MSM	Edit + warp operations	More stable than DTW in general; parameter-sensitive.
SBD	Normalised cross-correlation	Strong under global phase shift; weak under local warping.
IDK	Distributional kernel	May improve with length; may lose temporal order.

Different similarity measures encode different invariances; the best choice depends on the data-generating mechanism, particularly the noise regime, degree of misalignment, and effective sequence length.

4 Results

We present the experimental results in two parts. Section 4.1 reports the cross-dataset benchmark on 18 UCR datasets under clean conditions (Part 1). Section 4.2 presents the controlled perturbation study across noise, shift, and length dimensions (Part 2), including the circular-shift ablation and the IDK windowing supplementary experiment.

4.1 Part 1: Cross-Dataset Benchmark

Table 3 reports the mean ARI (averaged over 10 clustering seeds) for each dataset–measure combination. Across all 18 datasets, the overall mean ARI ranks the five measures as: SBD (0.371) $\approx$ DTW (0.368) $>$ MSM (0.341) $>$ ED (0.279) $>$ IDK (0.218). The corresponding NMI ranking is consistent: SBD (0.413) $\approx$ DTW (0.412) $>$ MSM (0.382) $>$ ED (0.317) $>$ IDK (0.249).

Table 3: Part 1 benchmark: Mean ARI ($\pm$ initialisation variance) across 18 UCR datasets. Best result per dataset in **bold**.

Dataset	ED	DTW	MSM	SBD	IDK
Chinatown	0.133	0.212	0.133	0.072	−0.003
SyntheticControl	0.348	**0.783**	0.555	0.552	0.000
MoteStrain	0.432	0.237	**0.586**	0.408	0.000
ECG200	0.188	0.049	0.131	**0.180**	−0.002
CBF	0.253	**0.665**	0.473	0.491	0.238
TwoPatterns	0.023	**0.658**	0.080	0.096	0.125
ECGFiveDays	0.004	0.024	0.331	**0.794**	0.068
Plane	0.614	**0.808**	0.722	0.670	0.766
GunPoint	−0.005	−0.005	−0.005	−0.005	**0.051**
Wine	−0.005	−0.003	−0.008	−0.007	**0.013**
ArrowHead	**0.283**	0.196	0.221	0.226	0.189
Trace	0.367	**0.645**	0.414	0.593	0.212
Coffee	**0.693**	0.530	0.626	0.663	0.539
DiatomSizeReduction	0.772	0.757	0.678	**0.781**	0.558
Symbols	0.646	**0.714**	0.687	0.676	0.602
OSULeaf	0.116	0.133	0.218	**0.270**	0.223
Computers	−0.002	**0.035**	0.031	0.031	0.009
ACSF1	0.167	0.181	0.262	0.182	**0.341**
Overall mean	0.279	0.368	0.341	**0.371**	0.218

A Friedman test over the 18 datasets yields $\chi^2 = 8.09$, $p = 0.088$, which does not reach significance at $\alpha = 0.05$. The average ranks are: DTW (2.44), SBD (2.61), MSM (2.83), ED (3.50), IDK (3.61). Although the overall ranking is not statistically significant on clean data, several patterns emerge:

- DTW and SBD achieve the highest average ARI, with DTW winning on 7 out of 18 datasets (predominantly those with temporal-alignment requirements such as CBF, TwoPatterns, and SyntheticControl).
- SBD excels on datasets where phase invariance is beneficial (ECGFiveDays, DiatomSizeReduction).
- IDK achieves the lowest overall mean, but uniquely wins on GunPoint, Wine, and ACSF1—datasets where distributional structure rather than precise alignment carries the discriminative signal.
- MSM demonstrates consistent mid-range performance without catastrophic failures.

The key finding from Part 1 is that *no single measure dominates on clean data*. The non-significant Friedman test confirms that paradigm-level differences require additional stress—introduced by our perturbation protocol—to become statistically resolvable.

4.2 Part 2: Perturbation Study

Part 2 applies three systematic perturbations to CBF, Trace, and ECG200. Each perturbation–level–measure–seed combination produces one ARI value; Friedman tests are computed across all 15 perturbation cells (3 datasets $\times$ 5 levels) per perturbation type.

4.2.1 Overall Statistical Comparison

Table 4 reports the Friedman test results for the three perturbation types. All three tests are highly significant ($p < 0.005$), demonstrating that perturbation-induced stress *does* resolve paradigm-level differences that were invisible on clean data.

Table 4: Friedman test results for Part 2 perturbation study (ARI). CD = 1.575 (Nemenyi, $\alpha = 0.05$, $k = 5$, $N = 15$).

Perturbation	χ^2	p-value	Average Rank					Best
			ED	DTW	MSM	SBD	IDK	
Noise	27.41	1.6×10^{-5}	4.00	2.80	2.47	**1.60**	4.13	SBD
Shift	20.91	3.3×10^{-4}	4.47	2.27	2.93	**2.13**	3.20	SBD
Length	15.63	3.6×10^{-3}	4.07	2.67	**2.00**	2.73	3.53	MSM

The critical difference (CD) for Nemenyi post-hoc comparison is 1.575. Applying this threshold:

- Under **noise**: SBD (rank 1.60) is significantly better than ED (4.00) and IDK (4.13); DTW and MSM occupy the middle ground.
- Under **shift**: SBD (2.13) and DTW (2.27) are significantly better than ED (4.47); IDK (3.20) is not significantly different from either group.
- Under **length**: MSM (2.00) is significantly better than ED (4.07) and IDK (3.53); SBD (2.73) and DTW (2.67) are statistically indistinguishable from MSM.

4.2.2 Noise Degradation

As the noise level ℓ increases from 0.0 to 0.8, the five measures exhibit distinct degradation trajectories (degradation curves are provided as supplementary figures):

Table 5: Mean ARI under noise perturbation (averaged over 3 datasets $\times$ 10 seeds).

ℓ	ED	DTW	MSM	SBD	IDK
0.0	0.225	0.415	0.342	0.440	0.222
0.1	0.221	0.412	0.332	**0.461**	0.259
0.2	0.224	0.402	0.345	**0.449**	0.265
0.4	0.223	0.368	0.342	**0.434**	0.258
0.8	0.241	0.313	0.277	**0.362**	0.177

SBD maintains the highest ARI at every noise level and shows the most gradual degradation. Notably, SBD at $\ell = 0.8$ (ARI = 0.362) still exceeds DTW's baseline performance on ECG200. DTW degrades monotonically, losing 24.5% of its baseline ARI by $\ell = 0.8$—consistent with its elastic warping path overfitting noise-induced jitter. ED remains approximately flat, reflecting its insensitivity to noise when the noise is i.i.d. (the expected ℓ_2 distance increases uniformly across all pairs, preserving relative ordering). IDK shows non-monotonic behaviour: a slight improvement at moderate noise ($\ell = 0.1$–0.4) followed by collapse at $\ell = 0.8$, suggesting that distributional features gain discriminative power from mild noise regularisation but saturate under extreme corruption.

4.2.3 Shift Degradation (Edge Padding)

Table 6 presents mean ARI under temporal shift with edge padding.

Table 6: Mean ARI under temporal shift (edge padding, averaged over 3 datasets $\times$ 10 seeds).

Shift (%L)	ED	DTW	MSM	SBD	IDK
0%	0.225	**0.415**	0.342	0.440	0.222
5%	0.209	**0.405**	0.318	0.407	0.224
10%	0.162	**0.353**	0.292	0.319	0.194
20%	0.107	**0.315**	0.172	0.290	0.187
30%	0.070	**0.246**	0.103	0.170	0.153

Under edge-padded shift, DTW emerges as the most robust measure, retaining 59% of its baseline ARI at 30% shift. This validates its elastic warping mechanism: the Sakoe–Chiba band ($w = 0.1L$) can accommodate moderate temporal displacements. SBD degrades more than expected (61% loss at 30% shift)—an observation explained by the edge-padding protocol. When shifts are padded rather than wrapped circularly, the cross-correlation peak is attenuated by the flat-valued boundary regions, reducing NCC magnitude. ED collapses most severely (−69%), confirming that lock-step comparison is catastrophically sensitive to misalignment. MSM also degrades substantially (−70%), indicating that its edit operations do not effectively compensate for large global displacements. IDK shows moderate degradation (−31%), outperforming ED and MSM at 30% shift despite its lower baseline.

4.2.4 Length Degradation

Table 7 presents mean ARI under symmetric truncation with FFT resampling.

MSM is the most robust measure under length reduction, achieving the best ARI at all truncation levels below $f = 1.0$. Its split and merge operations provide a natural mechanism for handling the spectral smoothing introduced by FFT resampling of truncated signals. SBD

Table 7: Mean ARI under length perturbation (averaged over 3 datasets × 10 seeds).

Fraction f	ED	DTW	MSM	SBD	IDK
1.00	0.225	0.415	0.342	**0.440**	0.222
0.90	0.219	0.349	**0.388**	0.425	0.248
0.75	0.212	0.353	**0.347**	0.348	0.242
0.50	0.242	0.328	**0.336**	0.251	0.268
0.25	0.213	0.258	**0.292**	0.242	0.153

shows the steepest decline: from 0.440 ($f = 1.0$) to 0.242 ($f = 0.25$), a 45% loss. This is expected because truncation removes high-frequency components that contribute to the NCC peak, blurring the cross-correlation landscape. DTW maintains moderate robustness (38% loss at $f = 0.25$), while ED remains relatively flat—again reflecting its insensitivity to signal transformations that preserve relative pairwise ordering. IDK shows variable behaviour, with its ARI actually increasing at intermediate truncation levels ($f = 0.50$, 0.75) before declining at extreme truncation.

4.2.5 Circular-Shift Ablation

To isolate the intrinsic shift-invariance properties of each measure, we repeat the misalignment experiment with circular wrapping instead of edge padding. Under circular shifts, vacated positions are filled by wrapping from the opposite end of the series.

Table 8: Mean ARI under circular shift (averaged over 3 datasets × 10 seeds). Degradation relative to baseline (0% shift) shown in parentheses.

Shift (%L)	ED	DTW	MSM	SBD	IDK
0%	0.225	0.415	0.342	0.440	0.222
5%	0.208 (−8%)	0.342 (−18%)	0.323 (−6%)	0.416 (−5%)	0.210 (−5%)
10%	0.154 (−31%)	0.303 (−27%)	0.274 (−20%)	0.348 (−21%)	0.234 (+6%)
20%	0.087 (−61%)	0.236 (−43%)	0.150 (−56%)	0.232 (−47%)	0.242 (+9%)
30%	0.041 (−82%)	0.170 (−59%)	0.079 (−77%)	0.136 (−69%)	0.212 (−5%)

The circular-shift results reveal two important findings:

Finding 1: SBD does not achieve zero degradation with the standard library implementation. Despite being defined as invariant under circular shifts, the `aeon` library's SBD implementation uses zero-padded linear cross-correlation (via FFT with $2L$-point transform) rather than true circular cross-correlation. This implementation choice introduces boundary artifacts that cause 69% degradation at 30% shift. A separate verification using true circular cross-correlation (direct FFT on L-point signals) confirms that SBD achieves *exact zero error* under circular shifts—validating the mathematical definition while exposing a theory–practice gap in standard implementations.

Finding 2: IDK exhibits near-zero degradation under circular shifts. IDK shows only 5% degradation at 30% shift, and actually *improves* slightly at intermediate levels (10–20%). This is an unexpected finding: the Isolation Distributional Kernel computes similarity based on subsequence-level distributional overlap, which is inherently invariant to global temporal translation. Unlike SBD's invariance (which is exact by definition but implementation-dependent), IDK's shift robustness is an *emergent property* of its distributional mechanism.

4.2.6 IDK Windowing Analysis

A supplementary experiment compares direct (full-series) IDK against windowed (sliding-subsequence) IDK to understand IDK's poor baseline performance.

Table 9: IDK windowing comparison: Direct vs. sliding-window IDK.

Dataset	Mode	Window	ARI
Plane	Direct	$L = 144$	-0.000
Plane	Sliding	$w = 7$ $(0.05L)$	**0.800**
Plane	Sliding	$w = 14$ $(0.10L)$	0.777
Plane	Sliding	$w = 29$ $(0.20L)$	0.722
Trace	Direct	$L = 275$	-0.000
Trace	Sliding	$w = 14$ $(0.05L)$	0.224
Trace	Sliding	$w = 28$ $(0.10L)$	0.196
Trace	Sliding	$w = 55$ $(0.20L)$	**0.288**
ECG200	Direct	$L = 96$	-0.005
ECG200	Sliding	$w = 5$ $(0.05L)$	0.047
ECG200	Sliding	$w = 10$ $(0.10L)$	0.117
ECG200	Sliding	$w = 19$ $(0.20L)$	**0.154**

Direct IDK produces ARI ≈ 0 across all three datasets, confirming that a single Isolation Kernel evaluation on the full time series is insufficient to capture class-discriminative structure. Windowed IDK achieves dramatically higher performance: ARI $= 0.80$ on Plane (compared to 0.67 for SBD and 0.81 for DTW on the same dataset in Part 1). This demonstrates that IDK's distributional paradigm *requires* subsequence-level decomposition to be effective—consistent with CTDS [10], which uses sliding-window mean maps as its default feature extraction.

The optimal window size varies by dataset: smaller windows $(0.05L)$ favour Plane (where short, distinctive shape fragments exist), while larger windows $(0.20L)$ favour Trace and ECG200 (where longer morphological patterns carry the discriminative signal).

5 Discussion

Our results demonstrate that perturbation-induced stress reveals paradigm-level differences that are invisible on clean benchmark data. We now synthesise the empirical findings into mechanistic explanations, discuss limitations, and derive practical recommendations.

5.1 Mechanism–Performance Alignment

Each measure's degradation profile can be predicted from its mathematical definition:

ED (lock-step). Euclidean distance shows minimal sensitivity to noise (the expected distance inflates uniformly across all pairs, preserving relative ordering) but collapses under shift (-82% at 30% circular shift). This is a direct consequence of point-wise comparison without alignment: even a single-sample shift introduces $O(L)$ mismatched comparisons. ED's flat response to length perturbation further confirms that FFT resampling preserves the relative ℓ_2 structure when all series undergo identical spectral smoothing.

DTW (elastic). DTW achieves the best baseline performance and is the most robust under *edge-padded* shift, validating its elastic warping mechanism. However, it shows the steepest noise degradation among the top measures (-24.5% at $\ell = 0.8$): the warping path overfits noise-induced jitter, finding spurious alignments that increase within-cluster variance. Under circular shift, DTW degrades substantially (-59%), because circular displacements exceed the Sakoe–Chiba band width ($w = 0.1L$), and the warping path cannot recover from global misalignment larger than its constraint window.

MSM (elastic/edit). MSM excels under length perturbation (rank 1 across all levels), which we attribute to its split and merge operations. When FFT resampling smooths high-frequency content, the resulting amplitude distortions are naturally absorbed by MSM's cost-based edit operations, which penalise amplitude changes directly rather than requiring perfect pointwise alignment. MSM's moderate sensitivity to shift (-77% circular, -70% edge-padded) indicates that its edit operations are locally effective but cannot compensate for large global displacements.

SBD (sliding). SBD achieves the best Friedman rank under both noise (1.60) and shift (2.13) perturbations. Its noise robustness derives from the normalised cross-correlation formulation: NCC is computed as an inner product of normalised sequences, inherently averaging out i.i.d. noise contributions across all lags. Its shift robustness arises from the max-over-lags formulation, which by definition searches for the optimal alignment position.

The circular-shift ablation reveals a critical nuance: SBD's theoretical circular-shift invariance is exact only when implemented as true circular cross-correlation. The standard `aeon` library uses zero-padded linear convolution (FFT on $2L$ points), which introduces boundary artifacts under large shifts. This theory–practice gap has implications for practitioners relying on library implementations without verifying their circular-invariance guarantees.

IDK (distributional). IDK's most striking result is its near-zero degradation under circular shift (-5% at 30%), an emergent property not previously documented. The Isolation Distributional Kernel computes similarity based on the distributional overlap of subsequence isolation paths. Because this distribution is computed over all subsequences regardless of their temporal position, circular translation merely permutes the subsequence ordering without altering the distribution—conferring *de facto* shift invariance.

However, IDK's overall performance remains the weakest among the five measures. The windowing experiment reveals why: direct (full-series) IDK produces ARI ≈ 0, indicating that a single isolation kernel evaluation on the complete time series cannot distinguish between classes. Windowed IDK recovers substantial performance (up to ARI $= 0.80$), suggesting that IDK's value lies in capturing local distributional signatures that must be aggregated through a sliding-window mechanism.

5.2 The Role of Perturbation Type in Measure Selection

Our results support a *perturbation-conditional* approach to measure selection:

1. **Well-aligned, low-noise data**: DTW or SBD provide the best performance, with DTW favoured when temporal warping is expected and SBD favoured when global phase differences dominate.
2. **Noisy signals**: SBD is the clear first choice; its NCC formulation provides built-in noise suppression. MSM is a strong second choice.
3. **Potential misalignment**: SBD (with verified circular implementation) for periodic signals; DTW for aperiodic signals with moderate shift ($< 10\%L$).
4. **Variable temporal resolution**: MSM provides the most stable performance under length reduction, followed by DTW.
5. **Permutation-invariant scenarios**: IDK (with appropriate windowing) is uniquely suited when temporal order is unreliable or irrelevant.

This conditional framework contrasts with the common practice of defaulting to DTW for all time-series tasks, and demonstrates that mechanism-aware measure selection can yield substantial performance gains under specific data conditions.

5.3 Limitations

Several limitations of this study should be noted:

1. **Dataset scope.** The perturbation study uses only three datasets (CBF, Trace, ECG200). While these cover synthetic and real signals with different class structures, generalisation to larger and more diverse archives requires further validation.

2. **Hyperparameter sensitivity.** All measures use fixed, literature-default hyperparameters. DTW's warping window ($w = 0.1L$) and MSM's edit cost ($c = 1.0$) may not be optimal for all perturbation regimes. A hyperparameter-sweep study would complement our paradigm-level comparison.

3. **IDK windowing.** The windowing experiment uses only three window sizes per dataset. A systematic window-size optimisation study would better characterise IDK's potential.

4. **Univariate only.** Our framework evaluates univariate time series; multivariate extensions may reveal different paradigm-level trade-offs.

5. **Perturbation independence.** We apply noise, shift, and length perturbations independently. Real-world data may exhibit combinations of these degradations, potentially interacting in non-additive ways.

5.4 Implications for Library Implementations

The SBD circular-invariance gap highlights a broader concern: standard time-series libraries may implement measures in ways that do not preserve their theoretical properties. We recommend that:

- Library authors document whether their SBD implementation uses circular or linear cross-correlation.

- Practitioners who rely on SBD's shift invariance should verify the implementation or use explicit circular-CC routines.

- Future benchmarks should distinguish between a measure's *mathematical properties* and its *implementation-specific behaviour*.

6 Conclusion

This paper presents a mechanism-oriented comparison of five time-series clustering similarity measures—ED, DTW, MSM, SBD, and IDK—spanning the lock-step, elastic, sliding, and distributional paradigms. Through a unified k-medoids pipeline with controlled perturbation experiments, we provide empirical evidence for four main conclusions:

1. **No universal winner exists.** On clean data, DTW and SBD achieve comparable performance (mean ARI ≈ 0.37) without statistically significant separation. Perturbation stress is required to resolve paradigm-level differences.

2. **Perturbation type determines optimal measure.** SBD is the most robust under noise and temporal shift (Friedman rank 1.60 and 2.13 respectively); MSM is the most robust under length reduction (rank 2.00). These results are highly significant ($p < 0.005$ for all three perturbation types).

3. **Invariance properties predict degradation.** Each measure's empirical robustness profile aligns with its mathematical invariance structure: SBD's cross-correlation resists noise and shift; DTW's elastic warping handles moderate misalignment; MSM's edit operations absorb spectral smoothing; IDK's distributional kernel is inherently shift-invariant.

4. **Implementation matters.** SBD's theoretical circular-shift invariance is violated by standard library implementations that use linear (zero-padded) cross-correlation. IDK requires sliding-window decomposition to be effective; direct application produces near-zero clustering quality.

These findings support a *mechanism-conditional* approach to similarity measure selection: rather than applying a single default measure universally, practitioners should characterise their data's noise regime, alignment reliability, and temporal resolution, then select the paradigm whose invariance properties match the dominant source of variation.

Reproducibility. All code, experimental scripts, and result CSVs are available in the project repository. Experiments use fixed random seeds with `numpy.random.SeedSequence` for bit-exact reproducibility.

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